Unsupervised Clustering and Differential Expression Analysis of Lung scRNA-seq Data

By: Farah Cisse

Introduction

This project explores the scRNA-seq data of lung tissue in healthy mice. The lung tissue was chosen because it is the most biologically rich, containing a wide variety of different cell types. Specifically, this report will focus on identifying the specific genes that drive the separation between the distinct cell populations within the lung and what those specific biological cell types represent. This focus is interesting because it will enable insight into how the lung balances the programming of all these different cell types and how the cells are organized within the tissue. Ultimately, this report will showcase the power of unsupervised learning to discover structure in high-dimensional scRNA-seq data

II. Dataset and Key Methods

The dataset used for this report is the lung count matrix from the scRNA-seq experiments conducted for the Tabula Muris project. The dataset has 1923 rows, representing the genes, and 23433 columns, representing the samples, before any pre-analysis filtering has been performed. The first method used was PCA for dimensionality reduction, to summarize the most important information from thousands of genes into a smaller number of informative axes. K-means clustering was then applied to group cells into clusters based on similarity. Lastly, differential expression (DE) was used to compare gene expression in one cluster versus other clusters. This method identifies the genes that best distinguish each cluster.

III. Results and Discussion

III.I PCA Plot w/ K-means Clustering

To explore transcriptional heterogeneity within the lung tissue, PCA was performed on the gene expression matrix to capture the major axes of variation. The PCA projection reveals some separation of cells along the first two principal components. K-means clustering is then applied, partitioning the dataset into six groups (Figure 1).

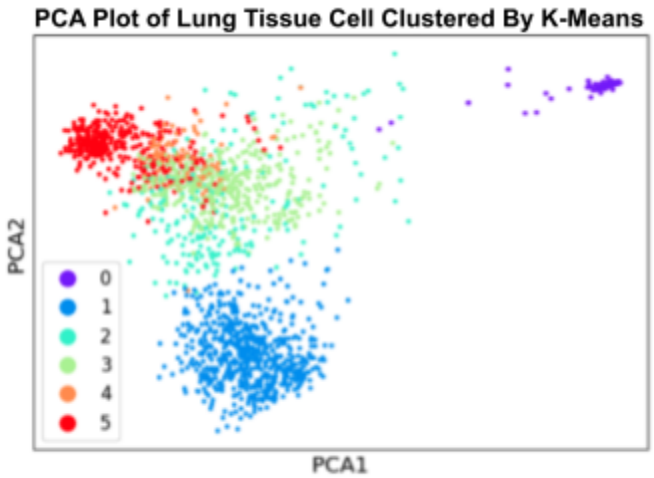

Figure 1: *PCA plot clustered using K-means clustering. Each point represents a cell and each distinct color is its own cluster.*

Clusters 0, 1, 2, and 5 show clear separation, with minor overlap between clusters 2 and 5. Clusters 3 and 4 show major overlap with clusters 2 and 5. The well-separated clusters likely indicate transcriptionally distinct cell populations with highly specialized, very different functions. In contrast, the overlapping clusters suggest closely related or transitional cell states. These cells may share gene expression profiles and have greater functional similarity. However, it's important to note that the observed overlap could be due to limitations with PCA. PCA captures only the largest sources of variation with linear structure. Therefore, some cell-type differences might only appear in later PC's that were not included, or in a non-linear structure that may not be captured by PCA. So, the clusters may overlap in a plot with PC1 and PC2 as the axes, but separate in higher dimensions. Overall, analysis strongly depends on the quality of the data and clustering, which can be influenced by technical noise and sparsity in the scRNA-seq data. As a result, rare cell types may be missed in this analysis.

III.II Top Differentially Expressed Genes in Each Cluster

After K-means clustering, DE analysis was performed to identify the genes that distinguish each cluster (Figure 2). Cluster 0 seemed to be dominated by housekeeping genes, present in many different types of cells, as well as ERCC spike-in genes, which are typically used as external controls in RNA-seq to correct for technical variability. This cluster overall lacked clear cell type specificity. Cluster 1 revealed a strong abundance for endothelial markers such as Cldn5, Esam, and Ramp2. These genes are consistent with the vascular endothelial cell type. Cluster 2 expressed immune-associated genes such as Srgn, Lsp1, and Cd53. This indicates an immune cell type. Cluster 3 showed enrichment for genes associated with mesenchymal or stromal identity. Cluster 4 had high expression for surfactant genes such as Sftpc and Sftpa1, matching it with the alveolar type II epithelial cell type. Lastly, Cluster 5 showed high expression of extracellular matrix and stromal genes such as Bgn and Loxl1.

Top 5 Differentially Expressed Genes Per K-Means Cluster

Cluster	0	1	2	3	4	5
rank_1	ERCC-00096	Ramp2	Srgn	Coro1a	Sftpc	Serping1
rank_2	Hsp90ab1	Cldn5	Lsp1	Crip2	Vim	Bgn
rank_3	ERCC-00130	Tspan7	Cd53	Laptm5	Sftpa1	Gsn
rank_4	ERCC-00002	Calcrl	Coro1a	Ptrf	Wfdc2	Sod3
rank_5	Tmbim6	Esam	Rac2	Cd52	Sftpd	Loxl1

Figure 2: *Table of top 5 genes in each K-means cluster that have the greatest differential expression. The columns are the K-means cluster number. The rows are the ranking from most differentially expressed genes (rank_1) to least differentially expressed genes (rank_2). Each entry contains a gene.*

Figure 2 only shows the top five differentially expressed genes per cluster. However, analysis of additional highly ranked genes revealed similar gene expression patterns. This reinforced the same biological cell type identities for each cluster. Relating these observations to the K-means clusters, well-separated clusters in the PCA space correspond to transcriptionally specialized cell types, such as endothelial and alveolar type II epithelial cells. Contrastingly, the overlapping clusters were more associated with related immune or stromal cells that share some of their transcriptional profiles. This reflects the continuous nature of the cell states throughout the lung. Despite these findings, it is important to acknowledge that the results found were based on gene expression alone. Additional validation, such as comparison with annotation datasets or experimental confirmation, should be performed to make more concrete conclusions.

III.III Validation

To validate the cluster identities, the k-means cluster assignments were compared to the annotations dataset created by the original scientists in the Tabula Muris project (Figure 3).

Comparison of K-means Clusters with Tabula Muris Lung Cell-Type Annotation

annotation cluster	lung endothelial cell	epithelial cell of lung	stromal cell	B cell	T cell	myeloid cell	monocyte	natural killer cell
0	0	0	3	0	0	0	0	0
1	0	0	1	0	0	0	0	0
2	688	1	11	0	1	2	0	0
3	2	11	1	57	52	80	65	37
4	1	96	1	0	0	0	0	0
5	2	3	383	0	0	0	0	0

Figure 3: *Table comparison of k-means cluster assignments with Tabula Muris lung cell-type annotations. Columns are the cell-types and rows are the k-means cluster. Values indicate the number of cells per cluster assigned to each annotated cell types.*

Based on the annotations, cluster 2 was dominated by lung endothelial cells, cluster 4 by epithelial cells of the lung, and cluster 5 by stromal cells. In contrast, cluster 3 contained a mixture of immune cell types, including B cells, T cells, monocytes, myeloid cells, and natural

killer cells. Clusters 0 and 1 lacked clear annotation and likely represent outliers or low-quality cells. As seen in the table, the annotation assignments didn't map perfectly onto the K-means clusters. This could be occurring for several reasons. First, the K-means cluster labels are arbitrary. The clusters identified by differential expression could appear under different cluster indices in the annotation table. Second, immune populations are transcriptionally similar and tend to cluster together, which explains why K-means grouped multiple immune cell types into a single heterogeneous cluster, while annotation distinguishes them into finer subtypes. Third, DE analysis identifies genes that differentiate clusters from one another but does not guarantee perfect alignment with cell type labels, especially when clusters contain mixed states. Despite this, the major biological conclusions are consistent across methods. One cluster represents endothelial cells, one represents stromal cells, another corresponds to alveolar type II epithelial cells, and one represents a mixture of immune cells. Minor discrepancies more so reflect methodological differences.